SPARKY'S SCIENCE EXPERIMENTS

Balloon Rocket on a String

Physical Science (forces and motion) · Ages 6+ · 20 minutes

Tape a balloon to a straw on a string and let it go - the air rushing out pushes the balloon along like a rocket.

Safety first

Sparky always says: Children under 8 need an adult with balloons - a burst piece can choke a small child!

- Throw burst balloon pieces away at once.

You need

- 3 to 5 metres of string or wool
- A drinking straw
- A balloon (long ones work best)
- Sticky tape
- Two chairs (or two trees)

Steps

1. Thread the string through the straw. Tie the ends to the chairs and pull it tight.
2. Blow up the balloon and pinch the end shut. Don't tie it.
3. Tape the balloon under the straw, with its opening pointing at one chair.
4. Pull it back to that chair, then let go. Whoosh!
5. Measure how far it went. Try less air, more air, or a sloping string.

If it doesn't work, try these in order

1. Doesn't move? Pull the string tighter and check the tape isn't sticking the straw to the string.
2. Spins round? Tape the balloon straight along the straw.
3. Stops early? Use more air or a smoother string.

Why it works

The balloon squeezes air out of the back, and the air pushes the balloon forward just as hard. Isaac Newton described this: for every action there is an equal and opposite reaction (his third law of motion). Real rockets work the same way, pushing hot gas out of the back very fast.

Did you know? In 1999, students at Stellenbosch University built SUNSAT, South Africa's first satellite.

Ask Sparky: What happens with a bigger balloon? Why does the balloon stop?